

Rutwik Patil

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SUMMARY

Results-driven **Data Analyst** with extensive **3+ years** of experience transforming raw data into actionable insights that enhance efficiency and streamline reporting processes. Proficient in **Python, SQL, and Power BI**, and automating workflows. Experienced in developing **ETL pipelines, predictive models, and dynamic dashboards** across various industries, including telecom, manufacturing, and EdTech. Skilled in utilizing **AWS and GCP**, and adept at collaborating cross-functionally to drive data-driven decisions and foster growth. **Actively seeking new opportunities** to contribute to innovative project

WORK EXPERIENCE

University of Maryland

Maryland, US

Data Analyst

Dec 2023 - May 2025

Project 1: Education Abroad Office

Addressed delays and lack of visibility in the university's manual course approval process, which relied on Google Drive and multiple advisor sign-offs. Automated the workflow using Salesforce, integrated internal databases, and implemented real-time tracking via a custom dashboard, reducing approval time from over a day to under an hour.

- Led end-to-end business analysis for automating the course approval process, reducing approval time from over a day to under an hour by designing **Salesforce** workflows and integrating internal databases
- Conducted stakeholder interviews and workshops to gather functional and non-functional requirements; documented findings through detailed **BRDs**, mockups, and process flows using Lucidchart
- Collaborated with cross-functional teams using **Agile methodologies**—organized tasks via Kanban boards, tracked progress with Gantt charts, and facilitated sprint planning to ensure timely delivery
- Developed and executed **User Acceptance Testing (UAT)** plans to validate system behavior, gather user feedback, and drive iterative improvements
- Utilized advanced **SQL** skills to write complex queries and construct procedures, functions, and workflows that supported application development, reporting, and data extraction processes
- Built a real-time administrative **dashboard** using **Salesforce Reports** and Dashboards to monitor approval statuses, highlight bottlenecks, and surface key student metrics
- Used **Power BI** to analyze historical student behavior trends, providing data-backed insights that influenced strategic decisions at the executive level

Project 2: Department of Decision Operations & Information Technologies

Drove adoption by developing practical AI solutions, guiding teams on data governance, and establishing the groundwork for future AI integration across the department.

- Developed an AI chatbot utilizing Google PaLM LLM and Langchain in **Python**, capable of transforming natural language prompts into **SQL** queries for efficient data retrieval from **MySQL**
- Advised project teams on implementing **data governance frameworks**, providing hands-on support in applying data quality standards to AI-driven solutions, and ensuring alignment with best practices
- Led hands-on sessions on **data quality best practices**, including **anomaly detection, data profiling, redundancy reduction, validation techniques, and compliance auditing** to ensure data integrity
- Organized and hosted a school-wide AI case competition, **authoring a case study** on Asha IT that explored the use of **AI to streamline hiring** and reduce HR costs for small businesses

Project 3: Center for Global Business

Resolved inefficiencies, formatting inconsistencies, and PII risks in student data reporting by replacing a legacy Excel-based system with automated data preparation workflows and dynamic Power BI dashboards, enabling real-time, FERPA-compliant reporting and improved decision-making.

- Automated data preparation workflows and validation checks significantly reduce manual effort and improve **data quality** for reporting.
- Created interactive **Power BI dashboards** using **DAX** for modeling, allowing stakeholders to visualize key metrics, monitor performance trends, and make data-driven decisions.
- Implemented advanced data masking and anonymization techniques to protect student PII, ensuring full compliance with **FERPA regulations** during analysis and dashboard development.
- Revamped the reporting process for the Center, earning formal recognition from the Dean's Office for driving **improved data visibility** and process efficiency.

Hughes Network Systems

Data Analyst

Remote, US

May 2024 - Aug 2024

Manual order placement and status tracking with vendors like AT&T and Comcast caused delays, frequent errors, and operational inefficiencies due to reliance on emails and Excel files. Developed a robust backend system using vendor APIs and engineered scalable data pipelines on Google Cloud Platform, automating the entire order lifecycle.

- Engineered scalable data pipelines on **Google Cloud Platform** using **Dataflow**, **BigQuery**, **Cloud functions**, and **AlloyDB**, reducing order processing times by 35%.
- Utilized advanced **SQL** skills to write complex queries and construct procedures, functions, and workflows that supported application development, reporting, and data extraction processes
- Leveraged expertise in **Python** to design and implement solutions, optimizing efficiency for a range of tasks and projects
- Designed an **AI-powered chatbot** integrated with a custom database to retrieve real-time order status, streamlining internal project management workflows and enhancing client communication.
- Developed a **regression model** to forecast order volume for a newly designed system, enabling better planning of required tools and resources for the project.
- Led **User Acceptance Testing (UAT)** and developed data-driven implementation roadmaps to validate solution performance, ensure stakeholder alignment, and mitigate operational risks.
- Collaborated with business stakeholders and the DevOps team, gathering requirements and creating Source-to-Target Mapping (**STTM**) documents to define **ETL processes** and ensure accurate data ingestion and transformation.
- Utilized **Microsoft Teams** and **Project** to track project progress and maintained comprehensive project documentation using **SharePoint** and **OneNote** for seamless collaboration and knowledge management.

Lechler

Data Analyst

Mumbai, India

Sep 2021 - Jun 2023

Fragmented data sources, manual reporting workflows, and limited visibility into quality and equipment metrics hindered root-cause analysis and proactive maintenance at Lechler. Built automated data pipelines and predictive analytics solutions to enhance data quality, accelerate reporting, and support operational decision-making.

- Developed advanced Power BI dashboards to monitor **production anomalies and quality deviations**, resulting in a 40% increase in the detection of manufacturing inefficiencies.
- Built an automated data cleansing pipeline using **Python (Libraries- pandas, numpy, plotly)** to detect and rectify errors (missing values, type mismatches, duplicates), reducing data prep time by 30%.
- Developed a **predictive model to forecast machine downtime** by analyzing operational data and failure patterns, resulting in a 15% improvement in reliability and enabling proactive maintenance strategies to minimize production disruptions.
- Optimized daily production data processing using **PySpark** on Databricks, increasing reporting speed by 21%—a critical improvement that enabled the development of real-time Power BI dashboards.
- Created a comprehensive **KPI catalog tool** mapping reported metrics to their respective dashboards, streamlining performance tracking, and improving stakeholder accessibility.
- Automated data ingestion from **AWS S3** using **Python** notebooks in Databricks, reducing manual intervention.

Mosdorfer

Data Analyst

Mumbai, India

May 2020 - Jul 2021

Siloed department-level data, inconsistent reporting formats, and outdated visualization methods at Mosdorfer led to inefficiencies in data consolidation and decision-making. Streamlined data operations by integrating performance data across departments and developing interactive dashboards to enable real-time visibility into key operational metrics.

- Designed and implemented an **ETL pipeline** to seamlessly process data from multiple branch locations and miscellaneous formats. Standardized data integration processes, reducing processing time by 20%.
- Created custom visual dashboards for performance tracking, contributing to improved communication with stakeholders, and a **10% increase in team operational efficiency**.
- Developed a **regression model** in Python to accurately forecast resource utilization, reducing prediction errors by 20% and improving schedule tracking efficiency by 23%.
- Developed an executive-level project progress dashboard consumed by the Board of Directors, earning **direct client recognition** for improving transparency and decision-making clarity.

- Led the establishment of comprehensive **data governance frameworks** and defined clear program goals, roles, and responsibilities. Instituted rigorous data standards, policies, and processes that enhanced data accuracy across the team.
- Led a cross-functional initiative to address rising labor costs by developing a dashboard to **analyze contractor and vendor spending**. Implemented strategic metrics that improved resource allocation, achieving a 9% reduction in labor expenses.

EDUCATION

University of Maryland MS in Information Systems (GPA: 3.7/4)	College Park, US Aug 2023 - Dec 2024
Narsee Monjee Institute of Management Studies B.E. in Electronics & Telecommunications	Mumbai, India Jul 2016 - Apr 2020

SKILLS

- **Databases:** Microsoft SQL Server, MySQL, MongoDB
- **Programming Languages:** Python (NumPy, Pandas, Matplotlib, Scikit-learn, PyTorch, NLTK, Plotly), R (Shiny, dplyr, ggplot2), Power Query, DAX
- **Tools & Libraries:** Excel, PowerPoint, Teams, MS Project, Visio, Trello, Figma, Kanban Board
- **Data Visualization:** PowerBI, Tableau, Looker, QuickSight
- **Big Data Tools:** Databricks, Alteryx, AWS Redshift, Google BigQuery, PySpark, Hadoop
- **Machine Learning:** Supervised ML (Linear and Logistic Regression, Decision Trees, RF, XGBoost), Unsupervised ML (K-means), NLP, Classification techniques (binary classification, multi-class classification), A/B testing, ROC-AUC Curve Analysis, Confusion Matrix

PROJECTS

- Reddit Data Engineering - Scalable ETL Pipeline with AWS & Airflow**
- Designed and optimized a scalable ETL pipeline using Python, Apache Airflow, and AWS S3 to process 50,000+ Reddit records, reducing execution time by 30% and boosting data accuracy by 15% through advanced cleaning, validation, and hypothesis testing—ultimately improving product success insights by 25%.
- AI-Driven Business Development Agent – AWS (Lambda, S3, CloudWatch, Redshift, IAM, OpenSearch)**
- Designed and deployed a multi-agent AI system on AWS, translating SQL queries into natural language outputs; developed data flow diagrams and transformation rules to define ingestion logic, track data lineage, and ensure consistency across AI-driven analytics, streamlining business development workflows and improving decision-making.
- Predicting Readmission Rates for Diabetic Patients**
- Leveraged PySpark on Databricks and advanced machine learning algorithms, including Random Forest, Regression, and XGBoost to analyze behavioral health data and develop predictive models, achieving 87% accuracy.
 - Utilized SMOTE to address data imbalance and developed predictive models that accurately predicted hospital readmission rates among diabetic patients with 87% accuracy.